Data Compression

If you're an ardent photographer you've probably noticed the option your camera offers of saving photographs in RAW format. That means you're using a format that retains the image in excruciating detail. Many formats have been developed considering the sensitivity of human perception to do away with excruciating detail that only go on to hog space in terms of MB and GB on your memory stick. To reduce the cost of storing images in

electronic memory certain standards like JPEG and MPEG were developed. JPEG, if you didn't know, is short for Joint Photographic Experts Group and MPEG stands for Moving Picture Experts Group.

What compression essentially does is loses superfluous information to trim the size of the file. How it does this is through different algorithms such as Lempel-Ziv, Huffman Coding and Discrete Cosine Transform to spot the repetitive and unnecessary details. For example the photograph of the sky would have hues of approximately the same shade. So the compression technique spots the pixels with approximately the same information (colour) and flattens this repetitive information by reducing the slight variations. This reduces the cost of storing the image because it occupies less space and because the lack of variations are not perceptible to the human eye the idea of compression is quite clever.